BE/APh161: Physical Biology of the Cell
Homework 2
Due Date: Wednesday, January 21, 2026

"Whatever you can do or dream you do, begin it. Boldness has genius, power and magic in it." - William Hutchison Murray in his 1951 book The Scottish Himalayan Expedition

This second problem set follows the same spirit of the first problem set by asking you to continue your skills as an order-of-magnitude thinker. As before, when doing street fighting estimates, the goal is to do simple arithmetic of the kind that all numbers take the values 1, few (f) or 10. few $\times$ few = 10, etc. Please do not provide estimates with multiple "significant" digits that are meaningless. Be thoughtful about what you know and what you don't know. You may use the Bionumbers website (http://bionumbers.hms.harvard.edu/) to find key numbers (examples are masses of amino acids (BNID 104877) and nucleotides (BNID 103828), the speed of the ribosome (BNID 100059), etc.), but please provide a citation to the Bionumber of interest as shown above. However, for many of these problems the essence of things is to do simple estimates, not to look quantities up. In particular, if in doubt, use the square root rule

$$x_{guess} = \sqrt{x_{low} x_{high}}, \tag{1}$$

which instructs us to take a lower and upper bound guess and then to take their geometric mean (which is the same as averaging their exponents).

1. Photons and your eyes: Seeing the North Star.

Polaris has been known to generations of northern hemisphere navigators as a tool for finding latitude by simple geometrical measurements with a sextant. In this problem you are asked to estimate the amount of light reaching our eyes from that famed star. The luminosity of Polaris is roughly 2000 times that of the Sun. Use that the solar irradiance at Earth is about 1000 W/m^2 and that Polaris is at a distance of 430 light years from Earth. For photon-counting estimates, you will need to adopt a typical visible photon energy (for example, corresponding to a wavelength of order 500–600 nm) and state what you chose.

(a) What is the power output of Polaris?

The sun has isotropic radiation. The Earth lies on the surface of a sphere of radius R_{SE}, where the measured solar flux is $\phi_S \simeq 10^3\,\mathrm{W\,m^{-2}}$. The total power emitted by the Sun is therefore

$$P_S = 4\pi R_{SE}^2 \phi_S. \tag{2}$$

Light from the Sun takes approximately 8 minutes to reach the Earth. Using the speed of light $c = 3 \times 10^8\,\mathrm{m\,s^{-1}}$, this gives

$$R_{SE} \simeq 8 \times 60 \times c \approx 1.4 \times 10^{11}\,\mathrm{m}. \tag{3}$$

Substituting into the expression above yields

$$P_S \approx 4\pi R_{SE}^2 \times 10^3 \sim 10^{26}\,\mathrm{W}. \tag{4}$$

The luminosity of Polaris is approximately 2000 times that of the Sun,

$$P_{\text{Polaris}} \sim 2000\,P_S \sim 2 \times 10^{29}.$$

(b) How many photons cross the pupil of your eye each second coming from this famous star and what is the corresponding mean spacing between photons along the line of propagation (that is, use the mean inter-arrival time and multiply by the speed of light)? For the pupil size, adopt and justify a diameter.

Polaris is located at a distance d from the Earth of approximately

$$d = 430\,\text{light-years} = 430 \times 365 \times 24 \times 3600 \times c \sim 4 \times 10^{18}\,\mathrm{m}. \tag{5}$$

The flux received at the Earth from Polaris is then

$$F = \frac{P_{\text{Polaris}}}{4\pi d^2}. \tag{6}$$

Using the estimate above gives

$$F \sim 10^{-9}\ W \cdot m^{-2}. \tag{7}$$

We take the pupil to have a radius of a few millimeters,

$$r_p \sim f\mathrm{mm}. \tag{8}$$

The corresponding pupil area is

$$A = \pi r_p^2 \sim 10^{-5}\,\mathrm{m}^2. \tag{9}$$

If the incident light has flux F, the optical power entering the eye is

$$P_{\text{eye}} = FA \sim 10^{-13}\,\mathrm{W}. \tag{10}$$

To estimate the photon arrival rate, we note that the energy of a single photon of wavelength λ is

$$E_\nu = \frac{hc}{\lambda} \sim 10^{-19}\,\mathrm{J}. \tag{11}$$

The corresponding photon flux entering the pupil is

$$\dot{N} = \frac{P_{\text{eye}}}{E_\nu} \sim 10^5\,\mathrm{s}^{-1}, \tag{12}$$

which implies a mean time interval between successive photons of order

$$\Delta t \sim 10^{-6}\,\mathrm{s}. \tag{13}$$

Along the direction of propagation, this corresponds to a typical photon spacing

$$\ell \sim c\,\Delta t \sim 1\,\mathrm{km}. \tag{14}$$

(c) What is the mean rate of arrival of photons to a single cone cell in the eye? State and justify the assumptions you need about how the star's image is distributed on the retina and the effective collecting area associated with a cone. Comment on what your answer implies about how vision works when photon arrivals are sparse. Your eye is not measuring a continuous intensity, it is counting rare events against noise and integrating over time.

Solution: We estimate the photon arrival rate to a single cone by (i) computing the diffraction-limited spot size of a point source (Polaris) on the retina, and (ii) estimating the effective cone "pixel size" from a crude human angular resolution. Take a dark-adapted pupil diameter $D \sim 5$ mm and a typical visible wavelength $\lambda \sim 550$ nm. The diffraction-limited angular scale is

$$\theta_{\text{diff}} \sim \frac{\lambda}{D} \sim \frac{5.5 \times 10^{-7}}{5 \times 10^{-3}} \sim 1 \times 10^{-4} \text{ rad.}$$

With eye focal length $f_{\text{eye}} \sim f$ cm, the linear spot radius/scale on the retina is

$$\Delta x_{\text{spot}} \sim f_{\text{eye}} \, \theta_{\text{diff}} \sim (f \times 10^{-2})(10^{-4}) \sim f \times 10^{-6} \text{ m} \sim f \ \mu\text{m}.$$

Thus the spot area is

$$A_{\text{spot}} \sim (\Delta x_{\text{spot}})^2 \sim (f \ \mu\text{m})^2 \sim 10 \ \mu\text{m}^2.$$

Assume the smallest resolvable feature is ~ 0.5 mm at 1 m, giving

$$\theta_{\text{eye}} \sim \frac{0.1 \text{ mm}}{1 \text{ m}} \sim 10^{-4} \text{ rad.}$$

The corresponding linear scale on the retina is

$$\Delta x_{\text{cone}} \sim f_{\text{eye}} \, \theta_{\text{eye}} \sim (f \times 10^{-2})(\times 10^{-4}) \sim f \times 10^{-6} \text{ m} \sim f \ \mu\text{m},$$

so an effective collecting area per cone is

$$A_{\text{cone}} \sim (\Delta x_{\text{cone}})^2 \sim (f \ \mu\text{m})^2 \sim 10 \ \mu\text{m}^2.$$

From part (b), the photon rate through the pupil is $\dot{N}_{\text{pupil}} \sim 10^5 \text{ s}^{-1}$. Let $\alpha \sim 0.1$ be an overall detection efficiency. If the starlight is distributed over the spot area, the mean detected rate per cone is

$$\dot{n}_{\text{cone}} \sim \alpha \, \dot{N}_{\text{pupil}} \frac{A_{\text{cone}}}{A_{\text{spot}}} \sim 10^4 \text{ s}^{-1}.$$

Although photon arrivals are fundamentally discrete events, a rate of $\sim 10^4 \text{ s}^{-1}$ corresponds to ~ 100 photons within a typical $\sim 10^{-2}$ s integration window of a cone. Thus, for Polaris, the visual system experiences a relatively smooth signal rather than extremely sparse photon counting. Truly sparse regimes occur only near visual detection threshold.
(Problem adapted from R. W. Rodieck's 1998 book *The First Steps in Seeing*)

2. Kelvin, Darwin and the Age of the Earth.

One of the great scientific debates of the 19th century was the age of the Earth and the Sun. In that time there was no knowledge of nuclear reactions, and the estimates based on known physics were very troubling for people like Darwin who assumed that life had existed on Earth for times much longer than these estimates would suggest. These estimates, most famously done by Lord Kelvin, also seemed to contradict estimates based on geological evidence, which suggested that the age of the Earth was more than a few hundred million years. This debate with Lord Kelvin really perturbed Darwin as shown in Figure 1.

(a) Assume that the Sun gets its power from burning something like coal. How long until the Sun burns out? Burning any ordinary chemical fuel will give the same order-of-magnitude estimate because chemical energies per unit mass are all of the same rough scale. Use your understanding of food to come up with your estimate of the energy density of fuel in units of J/kg. You will need the Sun's mass $M_\odot \approx 2\times10^{30}$ kg and its luminosity $L_\odot \approx 4\times10^{26}$ W.

Solution: A typical chemical energy density is set by food/combustion,

$$\epsilon_{\rm chem} \sim 10^7 \text{ J/kg}$$

For example, since a typical adult consumes ~ 2000 kcal per day and eats roughly 1 kg of food per day, the energy density of food is ~ 2000 kcal/kg $\approx 8\times10^6$ J/kg (using 1 kcal $\approx 4\times10^3$ J), which justifies taking $\sim 10^7$ J/kg as a street-fighting estimate. If the entire Sun of mass $M_\odot \simeq 2\times10^{30}$ kg were fuel, the available energy is

$$E \sim \epsilon_{\rm chem} M_\odot \sim 10^7 \times 2\times10^{30} \sim 2\times10^{37} \text{ J}.$$

With luminosity $L_\odot \simeq 4\times10^{26}$ W, the lifetime is

$$t \sim \frac{E}{L_\odot} \sim \frac{2\times10^{37}}{4\times10^{26}} \sim 5\times10^{10} \text{ s} \sim 10^3 \text{ yr}.$$

So chemical burning would power the Sun for only $\sim 10^3$ years.

(b) Assume that the Sun gets its energy from gravity, the release of gravitational potential energy as it contracts or assembles into a star. The key insight is that this energy can be written simply as

$$E_{\rm grav} = -\frac{3}{5}\frac{GM_{\rm star}^2}{R_{\rm star}}. \tag{15}$$

I love this calculation and if you want extra credit, you can derive it. Otherwise, feel free to use it as an off-the-shelf result. In light of this total energy, how long until the Sun burns out? (This is the Kelvin estimate.) State clearly what you assume for the Sun's radius $R_\odot$ and whether you take the available radiated energy to be $|E_{\rm grav}|$ or some fraction of it.

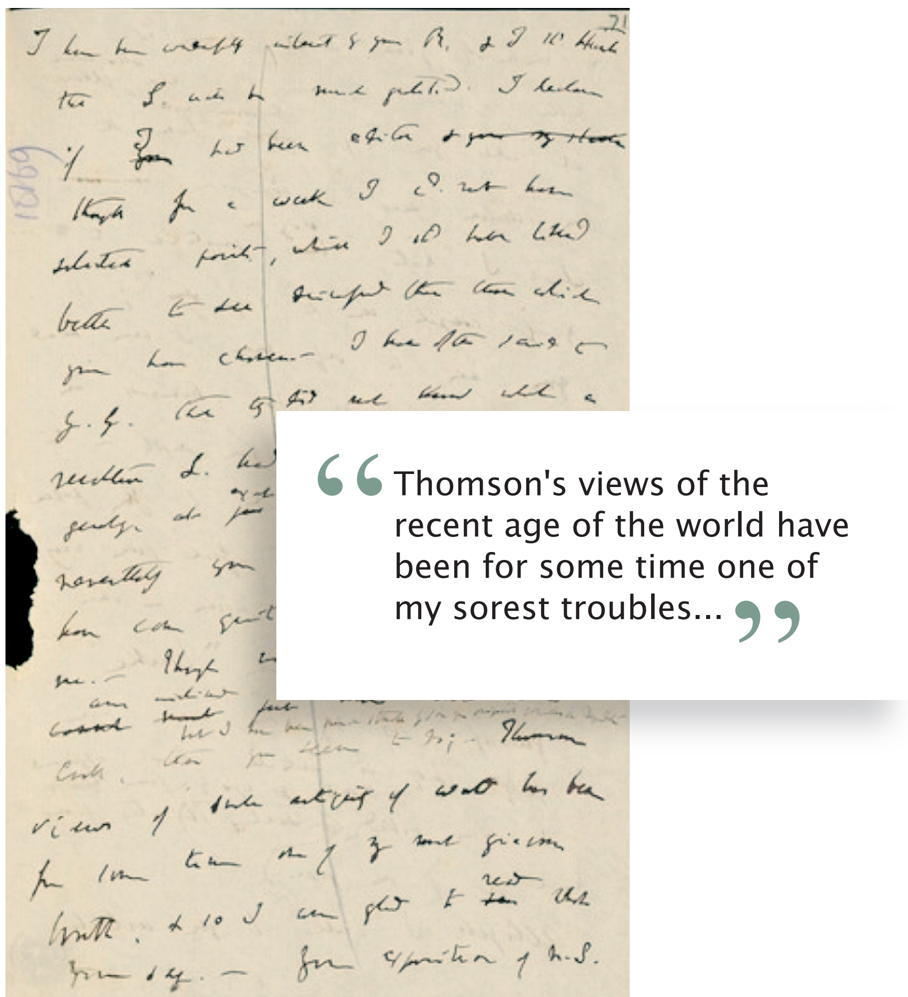

Figure 1: Letter from Charles Darwin to Alfred Russel Wallace in which he acknowledges the troubling implications of Lord Kelvin's estimates on the age of the Earth. Though the handwriting is nearly illegible to my eyes, the phrase "Thomson's views of..." can be found clearly about two-thirds of the way down. The transcription of the letter is given in the main text of this vignette. This letter, numbered DCP-LETT-6706, is housed in the Darwin Correspondence Project and dated 14 April 1869.

Solution: Take

$$R_\odot \sim 7 \times 10^8 \text{ m}, \qquad E_{\text{grav}} \approx -\frac{3}{5}\frac{GM_\odot^2}{R_\odot}.$$

Using $G \simeq 7 \times 10^{-11}$,

$$|E_{\text{grav}}| \sim \frac{GM_\odot^2}{R_\odot} \sim \frac{7 \times 10^{-11}\,(2 \times 10^{30})^2}{7 \times 10^8} \sim \frac{7 \times 10^{-11}\,4 \times 10^{60}}{7 \times 10^8} \sim 4 \times 10^{41} \text{ J}.$$

Taking the available radiated energy to be $\sim |E_{\text{grav}}|$,

$$t_{\text{Kelvin}} \sim \frac{|E_{\text{grav}}|}{L_\odot} \sim \frac{4 \times 10^{41}}{4 \times 10^{26}} \sim 10^{15} \text{ s} \sim 3 \times 10^7 \text{ yr}.$$

(c) Provide intuition in a paragraph or less as to why either Darwin or Kelvin could have an opinion about what constitutes a long time or short time. What kinds of evidence, timescales, or physical processes were most salient to each of them?

Solution: Darwin focused on slow biological and geological change: natural selection, erosion, sedimentation, and the accumulation of small variations, which suggested timescales of hundreds of millions of years or more. Kelvin emphasized energy accounting and heat/gravitation: given the observed solar luminosity and known physics (chemical or gravitational energy), the Sun could not shine for more than $\sim 10^7$ years. Each was extrapolating from the processes most quantitatively compelling in their domain.

3. RNA Polymerase and Rate of Transcription.

One of the ways in which we are trying to cultivate a "feeling for the organism" is by exploring the processes of the central dogma. Specifically, I want you to have a sense of the number of copies of the key molecular players in the central dogma as well as the rates at which they operate. Further, I argue that it is critical you have a sense of *how* we know these numbers.

(a) If RNA polymerase subunits β and β' together constitute approximately 0.5% of the total mass of protein in an *E. coli* cell, how many RNA polymerase molecules are there per cell, assuming each β and β' subunit within the cell is found in a complete RNA polymerase molecule? The subunits have a mass of 150 kDa each. State clearly what you use for the total protein mass per cell, for example, based on your estimate from Homework 1. (Adapted from problem 4.1 of Schleif, 1993.)

Solution: We have discussed measurements showing that the dry weight of an *E. coli.* cell is roughly 30% of its total mass, half of which is protein. The total mass of the cell is estimated to be a picogram from the assumption that its density is nearly that of water and its volume roughly 1 μm^3. It is given in the problem that the β and β' subunits together have a mass of 300 kDa and comprise 0.5% of the protein mass. The number of β, β' subunit pairs is assumed to equal the number of RNA polymerases (RNAP). Putting all this together yields

$$\# \text{ of RNAP} = \frac{\text{total mass of RNAP in cell}}{\text{mass per RNAP}} = \frac{0.15 \times 10^{-12} \text{ g} \times 0.005}{3 \times 10^5 \text{ Da} \times 1.6 \times 10^{-24} \text{ g/Da}} = 1.5 \times 10^3. \tag{16}$$

(b) Rifampin is an antibiotic used to treat *Mycobacterium* infections such as tuberculosis. It inhibits the initiation of transcription, but not the elongation of RNA transcripts. The time evolution of an *E. coli* ribosomal RNA (rRNA) operon after addition of rifampin is shown in Figures 2(A)–(C). An operon is a collection of genes transcribed as a single unit. Use the figure to estimate the rate of transcript elongation and report your answer in nucleotides per second. Use the beginning of the "Christmas-tree" morphology on the left of Figure 2(A) as the starting point for transcription.

Solution: Comparing Figure 1(A) and Figure 1(B), one sees that 40 seconds after rifampin addition roughly 1.5 kb of the DNA from the start site has become free of RNAP. The micrographs are aligned well enough that one can assume the left edge in all of them is the start site. Assuming that the last RNAP to initiate transcription did so at nearly the same time as rifampin addition, one can infer that this RNAP transcribed 1.5 kb of DNA in 40 seconds, implying an elongation rate of

$$\text{Elongation Rate} = \frac{1.5 \text{ kb}}{40 \text{ seconds}} \approx 0.04 \text{ kb/sec} \tag{17}$$

Making the same comparison of Figure 1(A) and 1(C), indicates an elongation rate of 3.5 kb/70 sec = 0.05 kb/sec, or roughly 50 nucleotides/sec.

(c) Using the calculated elongation rate estimate the frequency of initiation off of the rRNA operon. Make clear how you estimated the typical spacing between adjacent RNA polymerases along the operon and report your answer in initiations per second. These genes are amongst the most transcribed in *E. coli*.

Solution: The operon is roughly 6 kb long. Given the elongation rate in (b), one RNAP would require $\frac{6 \text{ kb}}{0.05 \text{ kb/sec}} = 120$ seconds to complete a transcript.
To estimate the rate at which transcripts of the operon are made, one needs the number of RNAP on the operon at any one time. Looking at the micrograph, one can make a rough

count that under normal conditions there are $10-20$ RNAP per kilobase and that the operon is roughly 6 kb long. This implies roughly

$$6 \text{ kb} \times 15 \text{ RNAP/kb} = 90 \text{ RNAP} \tag{18}$$

on the operon, and if each RNAP requires 120 seconds to complete transcription, then the initiation frequency of the operon is $90/120\text{s} = 0.75 s^{-1}$. This corresponds to a production rate that just over ~ 1 transcript per second.
Alternatively, we can notice that the mean spacing between RNAPs is roughly 1000nt/15 $\approx$ 67nt. The average speed of each such RNAP (from the previous part of the problem) is 50 nt/s. Hence, the initiation frequency is

$$\frac{50 \text{ nt/s}}{67 \text{ nt}} \approx 0.75 \text{ s}^{-1} \tag{19}$$

which is the same as our first result.

4. Composition of a cell.

Here we are going to do a rough atomic census of living material by thinking about the principal ingredients of a cell. To get a sense of the chemical makeup of the dry mass of a cell, we are going to focus only on proteins and nucleic acids. Each subpart requires assumptions on your part about cells and their molecular contents. State your assumptions explicitly (with units), give a one-sentence justification (either from memory, intuition, or a cited source such as Bionumbers / Cell Biology by the Numbers), and then carry them through consistently.

(a) Provide a simple and clean estimate for the volume and mass of a typical bacterium such as *E. coli*.

Solution: If we look at Fig. 2.1 in PBoC, we see that the shape of an E. coli bacterium can be approximated as a cylinder with two hemispheres on the two ends. Using the same figure, we can approximate the height of the cylinder as 1 μm, and the diameter of the hemisphere as 0.5μm. Then, we can calculate the volume as

$$V = \pi(0.5\mu m)^2 \times (1\mu m) + 4/3\pi(0.5\mu m)^3 = 1.309\mu m^3$$

For the mass, we can assume that E.coli is roughly the same density as water, 1 g/cm^3. It then follows that the mass is approximately 1 pg.

(b) Assume that 1/3 of the mass of a bacterium is dry mass and for simplicity, we ascribe all of that dry mass either to proteins or nucleic acids. We will take our elemental composition

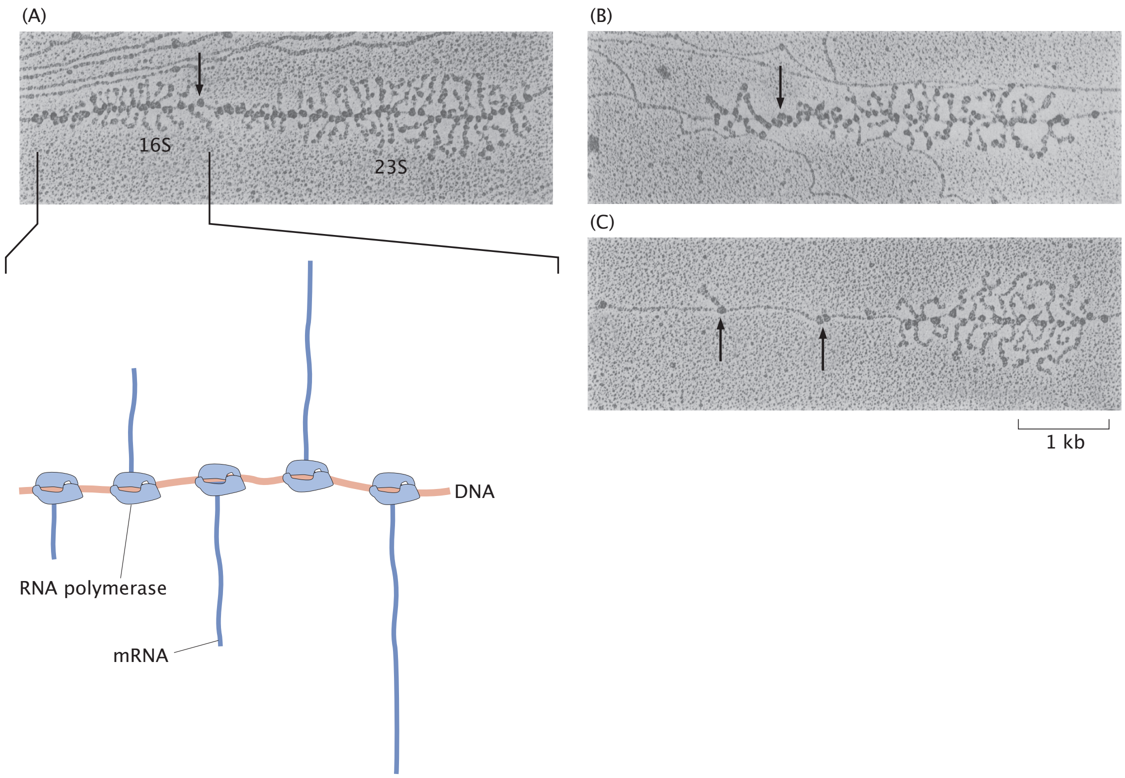

Figure 2: Effect of rifampin on transcription initiation. Electron micrographs of *E. coli* rRNA operons: (A) before adding rifampin, (B) 40 s after addition of rifampin, and (C) 70 s after exposure. No new transcripts have been initiated, but those already initiated are carrying on elongation. In parts (A) and (B) the arrow signifies the site where RNaseIII cleaves the nascent RNA molecule producing 16S and 23S ribosomal subunits. RNA polymerase molecules that have not been affected by the antibiotic are marked by the arrows in part (C). (Adapted from L. S. Gotta et al., *J. Bacteriol.* **173**(20), 6647–6649 (1991).)

of a "typical" amino acid to be $N_1C_5O_2H_8$ and a "typical" nucleotide to be $P_1N_5O_7C_{10}H_{14}$. Given that roughly half the dry mass of the cell is protein, work out the total protein mass in the cell, then estimate the number of protein molecules by adopting and defending a typical protein size (mass or length). From that, infer the number of amino acids per cell.

Solution: Using the elemental composition given for a "typical" amino acid, $\mathrm{N_1C_5O_2H_8}$, we estimate its molecular mass as $14 + 5 \times 12 + 2 \times 16 + 8 \times 1 \approx 114$ Da, which gives the typical molecular mass of ~ 100 Da. Since the total mass of all protein in the cell constitutes about 1/6 pg, we arrive at the total number of amino acids in a cell:

$$\frac{1}{6}\,\mathrm{pg} \cdot \frac{6 \times 10^{11}\,\mathrm{Da}}{\mathrm{pg}} \cdot \frac{\mathrm{aa}}{100\,\mathrm{Da}} = 10^9\,\mathrm{aa}. \tag{20}$$

Treating a "typical" protein as having 300 amino acids, these 10^9 amino acids correspond to few $\times\, 10^6$ proteins in our *E. coli* cell.

(c) As an alternative approach to estimating the total number of proteins in *E. coli*, assume that the bacterium is tightly packed with proteins (think of golf balls in a bathtub). How does this compare to the estimate from part (b)? If your two answers differ by more than an order of magnitude, identify which assumption is responsible.

Solution: If we model the bacterium as being tightly packed with proteins, we can estimate the number of proteins per cell by using protein volume. Assume the diameter of a typical (folded) protein is 5 nm $\approx f$ nm (from Bionumbers site) and that proteins are spherical. The volume of a typical sphere is then

$$V_{\mathrm{protein}} = \frac{4}{3}\pi(f\ \mathrm{nm})^3 \approx 100\,\mathrm{nm}^3. \tag{21}$$

Assuming the bacterium is tightly packed with proteins, we then estimate that there are about

$$\frac{1 \mu m^3}{100\,\mathrm{nm}^3} = 10^7 \text{ proteins per } \textit{E. coli}, \tag{22}$$

or more realistically slightly under given the efficiency of spherical tight packing. This result differs from the estimate in part (b) by roughly an order of magnitude, most likely arising from the assumption in (c) that protein packs the entire volume of the bacterium. Given that protein makes up a fraction of the dry mass, which is in turn a fraction of the cell mass, it is reasonable to expect the cell to not be tightly packed with protein throughout.

(d) Work out the number of nucleotides in the genome of our bacterium of interest. State clearly whether you estimated this from assumptions about genes or whether you looked up a genome size (and cite your source if you did).

Solution: Turning now to the number of nucleotides, we find that the molar mass of the "typical" nucleotide is around 300 Da, giving us a total number of nucleotides

$$\frac{1}{6}\,\text{pg} \times \frac{6 \times 10^{11}\ \text{Da}}{\text{pg}} \times \frac{\text{nt}}{300\ \text{Da}} \approx 3 \times 10^{8}\ \text{nt}. \tag{23}$$

We see that the number of nucleotides that are found in the genome (around 5 million) accounts for a negligible fraction of the entire nucleotide composition.

(e) Finally, figure out how many ribosomes are needed, translating at roughly 15 aa per second, to translate all of those proteins required to make a new cell within one division time. State and justify the division time you use. How many nucleotides are present in the ribosomal RNA making up all of these ribosomes?

Solution: We now turn to the source of non-genomic nucleotides, namely rRNA found in ribosomes. First, let's estimate the number of ribosomes that would be needed to synthesize the few million proteins discussed earlier. If we consider fast-growing cells, we have around 1200 seconds to create the billion peptide bonds between amino acids. This gives us a global synthesis rate of

$$\frac{10^{9}\text{aa}}{1200\text{s}} \approx 800,000\text{aa/s} \tag{24}$$

With each ribosome churning away at 15 aa/s, we need a total of 800,000/15 or 50,000 ribosomes to accommodate this global rate of protein production. As a sanity check, we see that this number meshes well with those provided in the previous problem.

Now to address how much of our nucleotide mass it tied up in these ribosomes, we refer to the total mass of a ribosome, 2700 kDa (BNID:100118), of which about 60% is RNA. This means that each ribosome has roughly 1600 kDa of nucleotides. Returning to the typical nucleotide mass of 300 Da, we arrive at

$$\frac{1600\ \text{kDa}}{\text{ribosome}} \times \frac{\text{nt}}{0.3\ \text{kDa}} = 5,000\ \text{nt/ribosome} \tag{25}$$

Multiplying by the total number of ribosomes, we arrive at 2.5×10^{8} total nt found in all ribosomes, which we see accounts for the vast majority of the total nucleotide compostion we found earlier.

(f) Given all of these numbers from the rest of this problem, you are now able to work out the overall composition of a cell. Provide an approximate formula for the stoichiometry of a bacterium.

Solution: This problem has divided up the bacterium into 2/3 wet mass (which we assume to be simply water) and 1/3 dry mass, which we have simplified to be described solely by proteins and nucleic acids. We must now determine the mol ratio among these components comprising the *E. coli* cell.

Given the total mass 1 pg estimated in part (a), and using the molar mass 18 g/mol for water (H_2O), there are

$$\frac{2}{3} \cdot 10^{-12}\, g \times \frac{1 \text{ mol}}{18 \text{ g}} \approx 40 \times 10^{-15} \text{ mol of wet mass (water).} \tag{26}$$

From part (c), we know there are 10^9 amino acids making up a cell, so there are

$$10^9 \cdot \frac{\text{mol}}{6 \times 10^{23}} \approx 2 \times 10^{-15} \text{ mol (aa).} \tag{27}$$

Finally, in part (e) we determined there are approximately 3×10^8 total nucleotides in the bacterium, so we have

$$3 \times 10^8 \cdot \frac{\text{mol}}{6 \times 10^{23}} = 0.5 \times 10^{-15} \text{ mol (nt).} \tag{28}$$

We now conclude that an approximate formula for the stoichiometry of a bacterium is

$$H_2O : \text{ aa} : \text{ nt} \approx 40 : 2 : 0.5 \equiv 80 : 4 : 1. \tag{29}$$

5. A feeling for the complete blood count (CBC) test.

Typical results for a complete blood count (CBC) are shown in Table 1. Assume that an adult has roughly 5 L of blood in his or her body. Choose either the male or female reference values (state which) and use street-fighting arithmetic. Based on these values estimate:

(a) The number of red blood cells. How does this compare to your estimate for the total number of cells in a human body and what does it tell you about how abundant red blood cells are?

Solution: From the table, we see that a typical red blood cell count is around 5×10^6 cells per μL. Scaling this up to the 5 L of blood, we get a total number of red blood cells,

$$5\text{L} \times \frac{10^6\ \mu\text{L}}{\text{L}} \times 5 \times 10^6 \frac{\text{cells}}{\mu\text{L}} \approx 2 \times 10^{13} \text{ cells}$$

(b) The percentage in volume they represent in blood. Do this in two ways: first using the hematocrit value, and second by combining the RBC count with the mean corpuscular volume (MCV). Comment on whether the two agree at the order-of-magnitude level.

Solution: Red blood cells have a volume of around 100 fL (BNID:110805). This means the total volume of all 20 trillion red blood cells is

$$100\frac{\text{fL}}{\text{cell}} \times 2 \times 10^{13} \times \frac{\text{L}}{10^{15}\text{fL}} = 2 \text{ L}$$

This means that the red blood cells take up two-fifths, or about 40% of the total blood volume. Coincidentally, the hematocrit value from the table corresponds to this value as empirically determined, and we see that 40% is right on the money for the actual value.

(c) Their mean spacing. State the simple geometric model you are using to convert number density into a spacing.

Solution: Converting our 5 L of blood into a length scale more meaningful for cells, we get that we have $5 \times 10^{15} \mu m^3$ of blood. Dividing by the number of cells, we get

$$\frac{5 \times 10^{15} \mu m^3 \text{blood volume}}{2 \times 10^{13} \text{cells}} = 250 \mu m^3$$

blood volume that each cell is "allotted". We can alternatively think of this as each cell getting an $\approx 6 \times 6 \times 6 \mu m$ box to call its own, meaning there is $\approx 6 \mu m$ spacing between cells.

(d) The total amount of hemoglobin in the blood.

Solution: Reading off the table, we see that hemoglobin is around $15 g/dL$. Scaling up to 5 L of blood, we get a total hemoglobin mass of

$$15\frac{\text{g}}{\text{dL}} \times 10\frac{\text{dL}}{\text{L}} \times 5L = 750\text{g}$$

(e) The number of hemoglobin molecules per red blood cell. State clearly how you convert from grams of hemoglobin to molecules.

Solution: To convert the mass of hemoglobin to a number of hemoglobin molecules we simply need to divide by the mass of a hemoglobin molecule. To estimate this, we harken back to Problem 2, where the typical amino acid is 100 Da and the typical protein, meaning the typical protein is around 30 kDa in mass. If we recall that hemoglobin is actually made of 4 subunits, we might adjust our estimate of hemoglobin mass to be four times larger, or around 120 kDa. (It turns out that this is a bit of an over estimate, since each subunit isn't as big as a "typical" protein, but this is sufficient for an order-of-magnitudes estimate). This gives us

$$\frac{750 \text{ g of hemoglobin}}{120 \text{ kDa}} \times \frac{6 \times 10^{20}\text{kDa}}{\text{g}} \approx 4 \times 10^{21}\text{hemoglobin molecules}$$

Finally, the number of hemoglobin per cell is

$$\frac{4 \times 10^{21}\text{hemoglobin}}{2 \times 10^{13}\text{cells}} = 2 \times 10^{8}\text{hemoglobin/cell}$$

(f) The number of white blood cells in the blood. As an optional extension, use the differential to estimate the counts of neutrophils and lymphocytes.

Solution: Again reading off the table, we see that a typical value for white blood cells is 8×10^3 per μL. Scaling up to 5 L of blood, we get a total number

$$5L \times \frac{10^6 \mu L}{L} \times 8 \times 10^3 \frac{\text{cells}}{\mu L} \approx 4 \times 10^{10}\text{cells}$$

(g) The number of platelets in the blood. Then, based on your answers, rank RBC, WBC, and platelets by abundance in blood and comment briefly on whether the most abundant "cell" in blood is also the most abundant cell type in the human body.

Solution: Taking a typical platelet count of $\sim 3 \times 10^5$ platelets/μL (the midpoint of the given range), the total number of platelets in 5 L of blood is

$$N_{\text{plt}} \approx 5 \text{ L} \times \frac{10^6\ \mu\text{L}}{\text{L}} \times 3 \times 10^5\ \frac{\text{platelets}}{\mu\text{L}} \approx 1.5 \times 10^{12} \text{ platelets.} \tag{30}$$

Test	Value
Red blood cell count (RBC)	Men: $\approx(4.3\text{–}5.7) \times 10^6$ cells/μL Women: $\approx(3.8\text{–}5.1) \times 10^6$ cells/μL
Hematocrit (HCT)	Men: $\approx$(39–49)% Women: $\approx$(35–45)%
Hemoglobin (HGB)	Men: $\approx$(13.5–17.5) g/dL Women: $\approx$(12.0–16.0) g/dL
Mean corpuscular hemoglobin (MCH)	$\approx$(26–34) pg/cell
MCH concentration (MCHC)	$\approx$(31–37)%
Mean corpuscular volume (MCV)	$\approx$(80–100) fL
White blood cell count (WBC)	$\approx(4.5\text{–}11) \times 10^3$ cells/μL
Differential (% of WBC):	
Neutrophils	$\approx$(57–67)
Lymphocytes	$\approx$(23–33)
Monocytes	$\approx$(3–7)
Eosinophils	$\approx$(1–3)
Basophils	$\approx$(0–1)
Platelets	$\approx(150\text{–}450) \times 10^3$ cell/μL

Table 1: Typical values from a CBC. (Adapted from R. W. Maxwell, Maxwell Quick Medical Reference, Tulsa, Maxwell Publishing Company, 2002.)

Using our previous estimates,

$$N_{\text{RBC}} \sim 2 \times 10^{13}, \tag{31}$$

$$N_{\text{WBC}} \sim 4 \times 10^{10}, \tag{32}$$

$$N_{\text{plt}} \sim 1.5 \times 10^{12}. \tag{33}$$

Ranking by abundance in blood gives

$$\text{RBC} > \text{platelets} > \text{WBC}. \tag{34}$$

Thus, red blood cells are the most abundant "cell" in blood. They are also among the most abundant cell types in the entire human body, indicating that a substantial fraction of all cells in the body are red blood cells.

6. Migration of the bar-tailed godwit

Animal migrations are one of the greatest of interdisciplinary subjects, bringing together diverse topics ranging from animal behavior to the physics of navigation to the metabolism required for sustained long-distance travel. The bar-tailed godwit is a small bird that each year travels between Alaska and New Zealand on the same kind of incredible nonstop voyage taken by happy tourists in modern long-distance jetliners as shown in Figure 3. During a visit to New Zealand's South Island, one of us had the chance to see these amazing birds in

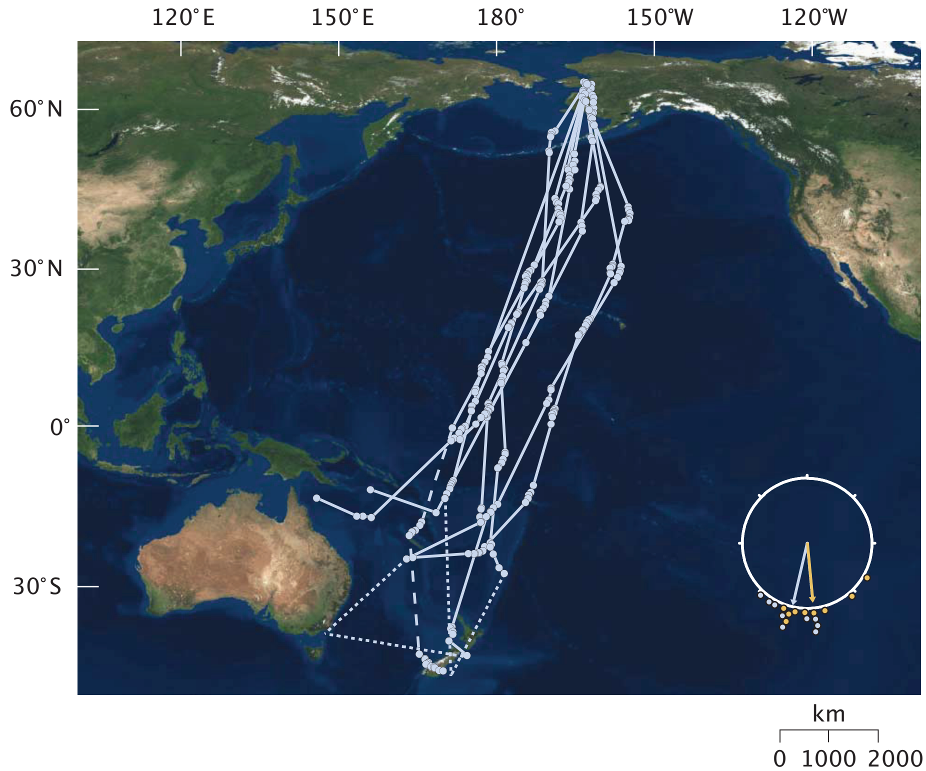

Figure 3: Map showing the migration pattern of the bar-tailed godwit. Adapted from Gill *et al.*, "Extreme endurance flights by landbirds crossing the Pacific Ocean: ecological corridor rather than barrier?", *Proc. Biol. Sci.* 276(1656), 447–457 (2009).

Okarito Lagoon with a naturalist guide who claimed that over the course of their ten-day, ten-thousand kilometer trip, these migratory birds lose 1/3 of their body mass. In this problem, we make a series of simple divide-and-conquer estimates to see whether this claim might be true. State and justify the numerical values you adopt for ρ, v, and L.

(a) Using dimensional-analysis arguments, work out how the drag force experienced by flying godwits depends upon the density of air (ρ), the speed of the birds (v) and the size of the birds (L). Specifically, work out the coefficients α, β and γ in the expression

$$F_{drag} = C\rho^{\alpha} v^{\beta} L^{\gamma}, \tag{35}$$

where C is a dimensionless constant that we will not consider further.

Solution: Because the units of force are kg$\cdot$ m/s^2, and the quantity ρ is the only parameter on the right of Eq. 35 that features the mass, we can conclude that $\alpha = 1$. Once we have determined α, the whole structure falls like dominoes. Using similar reasoning, we see that $\beta = 2$ since v is the only quantity featuring the time, and we need two powers of time in the denominator to match the units on the left side of the equation. Dimensional consistency then requires that $\gamma = 2$ as well, signaling that the drag force scales with the area of the moving object.

(b) Work out the power expended by the bar-tailed godwit to overcome the drag force. Then, work out the total energy expended during the ten-day migration in overcoming this drag force. State and justify how you convert from the 10,000 km and 10 day information to an estimate of the flight speed.

Solution: We can compute

$$\text{power during flight, } P = F_{drag} v = const. \cdot \rho v^3 L^2 \tag{36}$$

and the total energy expended is

$$\text{energy expended, } E = P t_{tot} \tag{37}$$

where t_{tot} is the total time of flight. For the purpose of an order of magnitude estimate, we will assume that the dimensionless constant in (36) is of order unity. The average velocity of the bird during its trip can be estimated as

$$v = \frac{\text{total migratory distance}}{\text{time of flight}} \tag{38}$$

$$\approx \frac{10^4 \text{ km}}{10 \text{ days}} \tag{39}$$

$$\approx \frac{10^7 \text{ m}}{10 \times 10^5 \text{ s}} \tag{40}$$

$$\approx 10 \text{ ms}^{-1} \tag{41}$$

By looking up pictures of satellite trackers being put on bar-tailed godwits, we can see that the bird is roughly the size of a human palm, i.e. $L \approx 10^{-1}$ m. The density of air is $\rho \approx 1$ kg m^{-3}. Then we have

$$P \approx (1 \text{ kgm}^{-3}) \times (10 \text{ ms}^{-1})^3 (10^{-1} \text{ m})^2 \approx 10W \tag{42}$$

and

$$E = P t_{tot} \approx (10 \text{ W}) \times (10 \times 10^5 \text{ s}) \approx 10^7 \text{ J} \tag{43}$$

(c) Given that burning fat yields 9 kcal/g, work out the number of grams of fat that would need to be burned to sustain the ten day flight of the bar-tailed godwit. Make clear what you assume about conversion efficiency between metabolic energy and mechanical work (for example, treat it as 100% for a lower bound, or adopt a plausible efficiency and state it). What fraction of the bird's body mass would be lost during such a migration based on these estimates, and how does it compare to the claimed 1/3 loss?

Solution: Using our result from the previous part,

$$\text{total fat required} = \frac{10^7 \text{ J}}{9 \text{ kcal/g}} \tag{44}$$

$$\approx \frac{10^7 \text{ J}}{9 \times \text{kcal/g} \times 4000\text{J/kcal}} \tag{45}$$

$$\approx 250 \text{ g} \tag{46}$$

Taking the birds to be spherical with the density of water, we can estimate a body mass of ~ 500 g. Therefore, the claim that the bar-tailed godwit loses 1/3 of its body mass over its migration is order-of-magnitude correct.

Note: Our estimate for the energy expenditure is an overestimate by a factor of a few - much of this error is due to the rough way we have estimated the drag force experienced by the bird, from which we should know to only expect order-of-magnitude results. The body mass of the bird is also slightly overestimated here, with bar-tailed godwits weighing in closer to ~ 300 g.

7. Real Estate for the Factories of ATP Synthesis

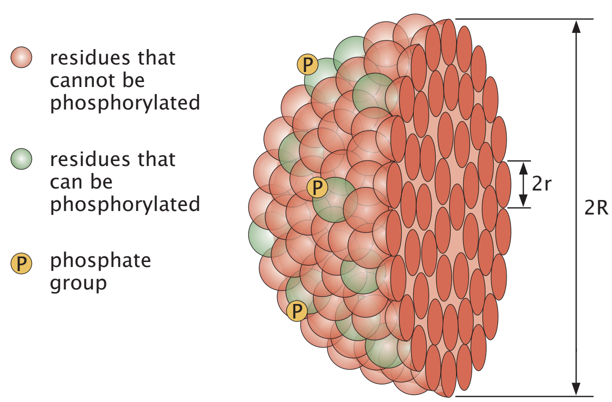

Figure 4: Schematic of a protein showing the surface residues that are available for phosphorylation.

We are captivated by the tension between those things about living organisms that are universal and those things that are baroque and specific to a given organism. One of the nearly universal features of living organisms on our planet is their use of ATP hydrolysis as an energy source for a huge variety of processes. A key empirical observation is that many cells operate with a roughly constant ATP demand per unit volume. Where does all of this ATP come from? Cells have tiny molecular machines known as ATP synthase in the membrane which use an ion gradient to drive the rotation of these machines to produce ATP. However, ATP is consumed throughout the volume of cells, but is produced on membranes. This leads to the possibility that as cells get bigger, there may be a point at which the surface area is insufficient to keep up with the demands of the cytoplasmic volume. Indeed, this problem explores the hypothesis that for cells above a certain size, ATP synthesis at the plasma membrane (such as in bacteria) no longer suffices and that a new specialized energy factory (i.e. the mitochondria) is required.

In what follows, keep careful track of two different quantities: a volumetric ATP demand density q with units of ATP/(μm^3 s) and a surface ATP production flux j with units of ATP/(μm^2 s).

(a) By considering the cost of protein synthesis for a dividing bacterium with a fast division time of 1000 s, justify the assertion that the volumetric ATP demand density is of order

$$q \approx 10^6 \frac{\text{ATP}}{\mu m^3\ s}. \tag{47}$$

State and justify the numerical values you adopt and comment briefly on whether protein synthesis is plausibly a dominant contribution to the ATP budget at the level of order-of-magnitude accuracy.

Solution: Here we use the estimate that one *E. coli* cell weighs about 1pg, 30% of which is dry mass and of which half is protein. This would give us a total of 0.1pg of protein weight. Using the fact that one amino acid weighs about 100 Daltons, and 1 Dalton equals 1g/mol, we can compute the number of amino acids to be 0.1pg $\times$ mol/100g $\approx 10^9$. Assuming that all these amino acids connected by polypeptide bonds, that is a total of 4×10^9 ATPs that are required to make all proteins in the cell. Given the division time of 1000s, rate of ATP consumption is 4×10^6ATP/s. To get the power usage, we use that *E. coli* has a volume of about 1μm^3, which gives us a power usage of

$$\text{power density} = 4 \times 10^6 \text{ATP}/\mu\text{m}^3 s.$$

(b) As shown in Figure 5, estimate the surface ATP production flux j supplied by ATP synthase on a membrane, in units of ATP/(μm^2 s). Your result will depend upon an areal density of ATP synthase (number per μm^2) multiplied by an ATP production rate per synthase (ATP/s). State your assumptions explicitly.

Solution:

Solution: We estimate

$$j \sim \rho_{\mathrm{ATPase}}\, k_{\mathrm{ATP}},$$

where ρ_{ATPase} is the areal density of ATP synthase (number/μm^2) and k_{ATP} is the ATP production rate per synthase. From Figure 5, each ATP synthase produces

$$k_{\mathrm{ATP}} \sim 10^2 \text{ ATP/s}.$$

To estimate the maximum packing density, we take an ATP synthase footprint of order

$$A_{\mathrm{prot}} \sim 100 \text{ nm}^2,$$

which corresponds to a protein tens of nanometers across. The maximal areal density is therefore

$$\rho_{\mathrm{ATPase}} \sim \frac{1}{A_{\mathrm{prot}}} \sim \frac{1}{100 \text{ nm}^2} = 10^{-2} \text{ nm}^{-2} = 10^4 \ \mu\text{m}^{-2},$$

since 1 μm^2 = 10^6 nm^2. Thus the maximal surface production flux is

$$j \sim (10^4 \ \mu\text{m}^{-2})(10^2 \text{ s}^{-1}) \sim 10^6 \ \frac{\text{ATP}}{\mu\text{m}^2\,\text{s}}.$$

Hence j is of order 10^6 ATP/(μm^2 s).

(c) Now consider a spherical cell of radius R whose cytoplasm demands ATP at volumetric rate density q and whose surface can supply ATP at flux j. The total ATP demand is $q(4\pi R^3/3)$ and the total ATP supply is $j(4\pi R^2)$. Use these to compute the maximum radius $R_{\max}$ such that surface ATP production can keep up with volume demand. Express your final result both as a formula in terms of j and q and as a numerical estimate using your values from parts (a) and (b). Comment on how eukaryotes evade this surface-area limitation.

Solution: Total demand and supply are

$$\dot{N}_{\mathrm{demand}} = q\left(\frac{4\pi R^3}{3}\right), \qquad \dot{N}_{\mathrm{supply}} = j(4\pi R^2).$$

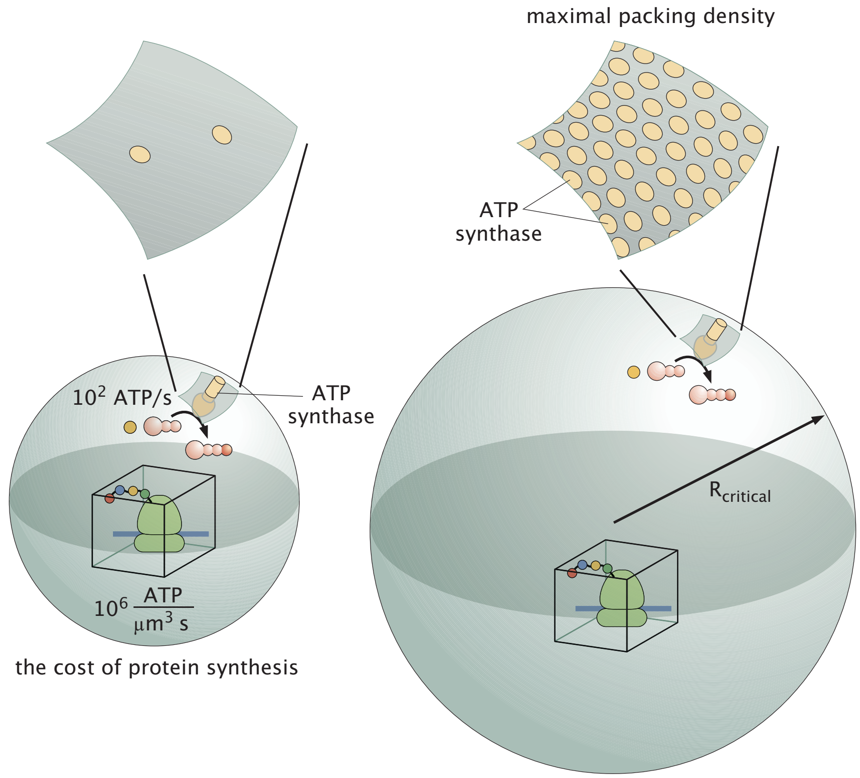

Figure 5: Surface coverage of bacterial cells with ATP synthase. For small cells, the demands of the cytoplasmic power consumption can be met by ATP synthases on the plasma membrane. However, for larger cells, there is not enough surface area to keep up with the demands of the power needs of the cellular interior.

Requiring $\dot{N}_{\text{supply}} \geq \dot{N}_{\text{demand}}$ gives

$$j(4\pi R^2) \geq q\left(\frac{4\pi R^3}{3}\right) \quad \Rightarrow \quad R \leq R_{\max} = \frac{3j}{q}.$$

Using $q \sim 10^6$ ATP/(μm^3 s) from part (a) and $j \sim 10^6$–10^7 ATP/(μm^2 s) from part (b),

$$R_{\max} \sim \frac{3j}{q} \sim 3\text{–}30\ \mu\text{m},$$

so cells much larger than a few–few×10 microns cannot meet ATP demand using only the plasma membrane. Eukaryotes evade this surface-area limitation by introducing mitochondria whose highly folded inner membranes provide much more membrane area (and thus more ATP synthase) per unit cell volume.